

VISHVAJIT S

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Education

Chennai Institute of Technology

B.E. Electrical and Electronics Engineering – CGPA 8.56/10

Chennai

2024 – 2028

Velammal Vidhyashram

Class XII (CBSE) 89.2% • Class X (CBSE) 92%

Chennai

2024 / 2022

Competitions and Leadership

National Champion, Unmanned Aerial Systems – IndiaSkills 2026 (NSDC)

2026

Advanced independently through state, regional, and national rounds

- Won the **national title** in Unmanned Aerial Systems, competing independently through three qualifying rounds; category carried a prize pool of **INR 1.75 lakh**.

Team Captain – ISRO IROC-U

2026

- Engineered a **zero-error docking mechanism** enabling a drone to dock and recharge autonomously irrespective of landing orientation, and a **GPS-denied drone** capable of independently executing a mission, landing, recharging, and resuming operation.

NIDAR 2026, Team Captain – national finalist; **Top 10**, Design Report

2026

NIDAR 2027, Captain – leading an indoor SLAM-enabled ROS 2 drone for disaster response

2026 – present

Aerathon, Avionics Lead and Pilot – national finalist both years; flew the manual mission

2025, 2026

ADDC, Vice Captain – **3rd place** nationally, Best Design Report

2025

Boeing Aeromodelling Challenge, Team Captain – regional finalist

2026

Projects

Fixed-Wing Aircraft – Independent Design and Development

XFLR5, Fusion 360, Ansys CFD

End-to-end aircraft design, fabrication, and flight test

- Designed airfoil and airframe geometry in XFLR5 and Fusion 360, validated aerodynamic performance using Ansys CFD, then fabricated and flight-tested the aircraft independently.

Hybrid VTOL Aircraft – Independent Build and Flight Test

Assembly, propulsion tuning, and transition flight testing

- Assembled and tuned a hybrid VTOL platform independently, achieving **stable transition** between vertical take-off/landing and fixed-wing forward flight.

PVC Rocker-Bogie Rover

Fusion 360, ROS 2 Jazzy, Raspberry Pi 5

1 m six-wheel rover – CAD design through hardware prototype

2026 – present

- Engineered a parametric rocker-bogie suspension in Fusion 360 targeting a **10 kg payload** at a bill-of-materials cost under **INR 50,000**, and a differential-bar linkage fabricable with standard hand tools; developing a ROS 2 skid_drive node on Raspberry Pi 5 converting /cmd_vel to skid-steer motor commands.

Experience

Centre for Drones and Space Research (CDSR), CIT Chennai

Chennai

Centre of Excellence for UAV and space research – developed and flight-tested all aircraft, drones, and the autonomous docking system above

2024 – Present

Technical Skills

- **Robotics:** ROS 2 (Jazzy), colcon, QoS, executors/callbacks, Gazebo, MAVROS, ArduPilot, GUIDED/offboard modes, SLAM
- **Languages:** Python, C, MATLAB (basics)
- **Embedded/MCU:** Teensy, ESP32/ESP8266, GPIO/PWM (pigpio, gpiozero)
- **Power electronics & PCB:** PCB design (KiCad, EasyEDA, Altium), buck/boost converters, power distribution, motor drive circuits
- **CAD/CFD/Simulation:** Fusion 360, XFLR5, Ansys CFD, 3D printing